Type MS · Estimating an integral using a series expansion

Answer 1 of 5 · Question 5 part d · 4 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

5 a	$(2+3x)^{-1}=\frac{1}{2}\left(1+\frac{3}{2}x\right)^{-1}$ so $p=\frac{1}{2}$ and $q=\frac{3}{2}$ oe	B1 B1 (2)
b	$\left(\frac{1}{2}\right)\left[1+(-1)\left("\frac{3}{2}"x\right)+\frac{(-1)(-2)}{2!}\left("\frac{3}{2}"x\right)^2+\frac{(-1)(-2)(-3)}{3!}\left("\frac{3}{2}"x\right)^3+\ldots\right]$	M1 A1ft
	$\frac{1}{2}-\frac{3}{4}x+\frac{9}{8}x^2-\frac{27}{16}x^3+\ldots$	A1 (3)
c	$\frac{1}{2}-\frac{3}{4}x+\frac{9}{8}x^2-\frac{27}{16}x^3+\frac{1}{2}x-\frac{3}{4}x^2+\frac{9}{8}x^3+\ldots$	M1
	$\frac{1}{2}-\frac{1}{4}x+\frac{3}{8}x^2-\frac{9}{16}x^3$	A1 (2)
d	$\int_0^{0.5}\left(\frac{1}{2}-\frac{1}{4}x+\frac{3}{8}x^2-\frac{9}{16}x^3\right)dx=\frac{1}{2}x-\frac{1}{8}x^2+\frac{1}{8}x^3-\frac{9}{64}x^4$	M1 A1
	$\frac{1}{2}\left(\frac{1}{2}\right)-\frac{1}{8}\left(\frac{1}{2}\right)^2+\frac{1}{8}\left(\frac{1}{2}\right)^3-\frac{9}{64}\left(\frac{1}{2}\right)^4=0.2256$ for awrt 0.2256	M1 A1 (4)
		Total 11 marks

Type MS · Estimating an integral using a series expansion

Answer 1 of 5 · Question 5 part d · 4 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

Part	Marks	Notes
(a)	B1	For $p=\frac{1}{2}$ or 2^{-1} or $q=\frac{3}{2}$
	B1	For $p=\frac{1}{2}$ or 2^{-1} and $q=\frac{3}{2}$
		NB $\frac{1}{2}\left(1+\frac{3}{2}x\right)^{-1}$ scores B1 B1
(b)	M1	For an attempt to expand $(1+qx)^{-1}$ with their value of q up to the term in x^3 It is not necessary to see p at this stage. The definition of an attempt is as follows: • The first term must be 1 • The next term must be correct for their value of q • The powers of qx must be correct eg $(qx)^2$ • The denominators must be correct Simplification not required. Do not allow missing brackets unless recovered later – this is a general point of marking.
	A1ft	For at least 3 terms fully correct and unsimplified for their value of q. It is not necessary to see p at this point.
	A1	For all 4 terms correct, all simplified.
	If there are any other methods used – send this to review please.	
(c)	M1	For multiplying their expansion by $(1+x)$ There must be a clear attempt to multiply to get 7 terms, allow up to 2 errors. Ignore terms with powers higher than 3.
	A1	For all 4 terms correct, ignore terms with powers higher than 3.
(d)	M1	For a minimally acceptable attempt to integrate an expression with at least 3 terms, which must include a term in x^3 See general guidance for definition of minimally acceptable attempt.
	A1ft	For a fully correct integration of their expression (defined as above for the M mark).
	M1	For substitution of limits into a changed expression, the correct way round – this can be implied if the final answer is correct. We don't need to see 0 substituted in.
	A1	For awrt 0.2256 (Note the calculator value is 0.2288461……)

Type MS · Estimating an integral using a series expansion

Answer 1 of 5 · Question 5 part d · 4 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

Question number	Scheme								Marks
	0	*0.25*	*0.5*	**1**	*1.5*	**2**	3		B2

Type MS · Estimating an integral using a series expansion

Answer 2 of 5 · Question 9 part e · 5 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 3 · find part e in this mark scheme extract

9(a)	$(1+2x)^{-\frac{1}{3}} = 1+\left(-\frac{1}{3}\right)(2x)+\frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)(2x)^2}{2!}+\frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)\left(-\frac{7}{3}\right)(2x)^3}{3!}+\ldots$ $=1-\frac{2x}{3}+\frac{8x^2}{9}-\frac{112x^3}{81}$	M1 A1A1 [3]
(b)	$-\frac{1}{2} < x \leqslant \frac{1}{2}$ accept $-\frac{1}{2} < x < \frac{1}{2}$	B1 [1]
(c)	$f(x) = (2+kx^2)\left(1-\frac{2x}{3}+\frac{8x^2}{9}-\frac{112x^3}{81}\right)$ $=2-\frac{4x}{3}+\frac{16}{9}x^2+kx^2-\frac{224}{81}x^3-\frac{2k}{3}x^3$	M1 M1A1 [3]
(d)	$-\frac{224}{81}-\frac{2k}{3}=-\frac{8}{3} \Rightarrow k=-\frac{4}{27}$	M1A1 [2]
(e)	$\int_{0.1}^{0.2}\left(2-\frac{4x}{3}+\frac{44x^2}{27}-\frac{8}{3}x^3\right)dx=\left[2x-\frac{4x^2}{2\times3}+\frac{44x^3}{3\times27}-\frac{8x^4}{4\times3}\right]_{0.1}^{0.2}$ $=\left(2\times0.2-\frac{2\times0.2^2}{3}+\frac{44\times0.2^3}{81}-\frac{2}{3}\times0.2^4\right)-\left(2\times0.1-\frac{2\times0.1^2}{3}+\frac{44\times0.1^3}{81}-\frac{2}{3}\times0.1^4\right)$ $=0.1828$ NB: Calculator value is 0.18301744	B1FT M1A1ft M1 A1 [5]
		Total 14 marks

Type MS · Estimating an integral using a series expansion

Answer 2 of 5 · Question 9 part e · 5 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 3 · find part e in this mark scheme extract

Part	Mark	Notes
(a)	M1	For an attempt at binomial expansion. • The first term is 1 • The denominators are correct. • The powers of x are correct.
	A1	For at least one algebraic term correct and simplified.
	A1	For a fully correct and simplified expansion.
(b)	B1	For the correct range. Accept $\|x\| < \frac{1}{2}$ Do not accept $-\frac{1}{2} \leq x \leq \frac{1}{2}$
(c)	M1	For showing the intent to multiply their expansion by $\left(2+kx^2\right)$
	M1	For expanding the two brackets up to terms in x^3. Ignore terms in higher powers.
	A1	For the correct expansion. This does not have to be simplified. Must be in ascending powers, but accept $\frac{16}{9}x^2$ and kx^2 in either order and accept $-\frac{224}{81}x^3$ and $-\frac{2k}{3}x^3$ in either order. $f(x) = 2 - \frac{4x}{3} + x^2\left(\frac{16}{9}+k\right) + x^3\left(-\frac{224}{81}-\frac{2k}{3}\right)$ ISW errors in collection of coefficients once correct expansion in ascending powers seen.
(d)	M1	For setting their coefficient of x^3 equal to $-\frac{8}{3}$
	A1	For the correct value of k
(e)	B1FT	For substituting in their k into coefficient for x^2 in their expansion from (c) and having the coefficient of x^3 as $-\frac{8}{3}$. If the coefficient of x^2 is not correct then the substitution must be seen to award this mark. Note: this is an M mark in epen.
	M1	For an attempt to integrate their expansion which must have 4 terms. At least two powers of x to increase by 1 (including $c \rightarrow cx$). No power of x to reduce.
	A1ft	For a correctly integrated expression follow through their 4 term polynomial. Simplified integrated expression is $2x - \frac{2x^2}{3} + \frac{44x^3}{81} - \frac{2}{3}x^4$
	M1	For substituting in 0.2 and 0.1 the correct way around in their integrated expression. Expression should not have been differentiated and needs to have changed. Not dependent on first M. Must show the substitution if their integral or their limits are incorrect. Correct awrt 0.1828 following correct integral with correct limits implies this mark.
	A1	For the correct value of awrt 0.1828 NB: The calculator value is 0.18301744

Type MS · Estimating an integral using a series expansion

Answer 2 of 5 · Question 9 part e · 5 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 3 · find part e in this mark scheme extract

Question	Scheme	Marks

Type MS · Estimating an integral using a series expansion

Answer 3 of 5 · Question 10 part d · 4 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part d in this mark scheme extract

10 (a)	$\frac{3}{\sqrt{9-3x}}=\frac{1}{\sqrt{9}}\times\frac{3}{\sqrt{1-\frac{3x}{9}}}=\frac{1}{\sqrt{1-\frac{x}{3}}}=\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$ *	B1B1* cso [2]
(b)	$\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}=1+\left(-\frac{1}{2}\right)\left(-\frac{x}{3}\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{x}{3}\right)^2}{2!}+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)\left(-\frac{x}{3}\right)^3}{3!}$ $\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}=1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}$	M1 A1A1 [3]
(c)	$3\mathrm{f}(x)=\frac{3(1+2x)}{\sqrt{9-3x}}=\frac{3(1+2x)}{3\sqrt{1-\frac{x}{3}}}=(1+2x)\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$ $3\mathrm{f}(x)=(1+2x)\left(1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}\right)$ $3\mathrm{f}(x)=(1+2x)\left(1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}\right)=1+\frac{13}{6}x+\frac{3}{8}x^2+\frac{41}{432}x^3$	B1 M1 M1A1 [4]
(d)	$\int_{0.1}^{0.2}\frac{3+6x}{\sqrt{9-3x}}\,\mathrm{d}x=\int_{0.1}^{0.2}\left(1+\frac{13}{6}x+\frac{3}{8}x^2+\frac{41}{432}x^3\right)\mathrm{d}x$ $=\left[x+\frac{13}{12}x^2+\frac{3}{24}x^3+\frac{41}{1728}x^4\right]_{0.1}^{0.2}$ $\int_{0.1}^{0.2}\frac{1+2x}{\sqrt{9-3x}}\,\mathrm{d}x=\frac{1}{3}\left[\left(0.2+\frac{13}{12}\times0.2^2+\frac{3}{24}\times0.2^3+\frac{41}{1728}\times0.2^4\right)-\left(0.1+\frac{13}{12}\times0.1^2+\frac{3}{24}\times0.1^3+\frac{41}{1728}\times0.1^4\right)\right]=0.044\ 470\ 2$	M1A1ft dM1A1 [4]
		Total 13 marks

Type MS · Estimating an integral using a series expansion

Answer 3 of 5 · Question 10 part d · 4 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part d in this mark scheme extract

	B1*	For simplifying to the required form $\frac{3}{\sqrt{9-3x}}=\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$
(b)	**M1**	For an attempt to find the binomial expansion for the given expression. • The expansion must begin with 1 • The denominators must be correct (ie. 2! And 3!) on the third and fourth terms. • The power of x must be correct (ie. $\left(-\frac{x}{3}\right)$ $\left(-\frac{x}{3}\right)^2$ and $\left(-\frac{x}{3}\right)^3$ with the correct corresponding denominators.
	A1	For one algebraic term correct and simplified
	A1	For the whole expansion correct up to x^3 Ignore extra terms beyond these. $\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}=1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}$
(c)	**B1**	For writing down $3f(x)=(1+2x)\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$
	M1	For replacing $\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$ with their $\left(1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}\right)$ provided their expansion has at least two terms one of which is an algebraic term.
	M1	For an attempt to expand $(1+2x)\left('1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}'\right)$ up to and including the term in x^3. Accept the expansion of $(3+6x)\left('1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}'\right)$
	A1	For the correct expansion $3f(x)=1+\frac{13}{6}x+\frac{3}{8}x^2+\frac{41}{432}x^3$ ignore extra terms
(d)	**M1**	For an attempt to integrate their expansion of 3f(x) or f(x) which must have as a minimum two terms, one of which is a term in x, x^2 or x^3 Ignore limits for this mark. They may multiply by $\frac{1}{3}$ at this stage so accept e.g $\left[\frac{x}{3}+\frac{13}{36}x^2+\frac{3}{72}x^3+\frac{41}{5184}x^4\right]$
	A1ft	For a correct integrated expression ft their expression.
	dM1	For substituting both values of 0.2 and 0.1 the correct way around seen. This mark can be implied by the value of 0.044 470 2 seen. If their expansion from (c) has not been divided by 3 for the integration, it must be divided by 3 here for the award of this mark.
	A1	For the correct value of 0.044 470 2 **NB**: The calculator value is 0.044 470 77

Type MS · Estimating an integral using a series expansion

Answer 3 of 5 · Question 10 part d · 4 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Estimating an integral using a series expansion

Answer 4 of 5 · Question 7 part e · 3 marks

2023 JUN · Paper 02 · extract 1 of 3 · find part e in this mark scheme extract

7(a)	$\left(1+\frac{x}{3}\right)^{-3}=\left[1+(-3)\left(\frac{x}{3}\right)+\frac{(-3)(-3-1)}{2!}\left(\frac{x}{3}\right)^2+\frac{(-3)(-3-1)(-3-2)}{3!}\left(\frac{x}{3}\right)^3\ldots\right]$ $=1-x+\frac{2}{3}x^2-\frac{10}{27}x^3$	M1A1 A1 [3]
(b)	$-3<x<3$ or $\lvert x\rvert<3$	B1 [1]
(c)	$\left((3+x)^{-3}=3^{-3}\times\left(1+\frac{x}{3}\right)^{-3}=\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}\right)$ $P=\frac{1}{27}$, $Q=\frac{1}{3}$	 B1, B1 [2]
(d)	$\frac{(1+4x)}{(3+x)^3}=(1+4x)\times\frac{1}{27}\times\left(1-x+\frac{2}{3}x^2\right)$ $=\frac{1}{27}\times\left(1-x+\frac{2}{3}x^2+4x-4x^2+\ldots.\right)$ $=\frac{1}{27}\left(1+3x-\frac{10x^2}{3}\right)$ or $\frac{1}{27}+\frac{x}{9}-\frac{10x^2}{81}$	 M1 A1 [2]
(e)	$\int_0^{0.2}\left(\frac{1}{27}+\frac{x}{9}-\frac{10x^2}{81}\right)\mathrm{d}x=\left[\frac{x}{27}+\frac{x^2}{18}-\frac{10x^3}{243}\right]_0^{0.2}$ $=\left(\frac{0.2}{27}+\frac{0.2^2}{18}-\frac{10\times0.2^3}{243}\right)-[0]$ $=0.0093004$	M1 M1 A1 [3]
		Total 11 marks

Type MS · Estimating an integral using a series expansion

Answer 4 of 5 · Question 7 part e · 3 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part e in this mark scheme extract

Part	Mark	Notes
(a)	M1	For applying a correct binomial expansion in unsimplified form. Minimum required: • The expansion begins with 1 • The next term is correct • The powers of $\frac{x}{3}$ must be correct eg $\left(\frac{x}{3}\right)^2$ • The denominators are correct. Do not allow missing brackets unless recovered later – this is a general point of marking. Ignore any terms with powers higher than 3.
	A1	Following M1 (this is a general point of marking, A marks can only follow M marks), all conditions above met, must see $1 - x$ and at least the term in x^2 or x^3 correct and simplified. Ignore any terms with powers higher than 3.
	A1	A fully correct and simplified expansion. Ignore any terms with powers higher than 3..
(b)	B1	For the correct validity.
(c)	B1	For the correct value of P or Q explicitly written or embedded in $\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}$
	B1	For the correct values of P and Q explicitly written or embedded in $\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}$
(d)	M1	For attempting to multiply their expansion, which must be of the form "P"× (their expansion from part (a), with a minimum of 3 terms, by $(1 + 4x)$. An attempt must include 3 correctly multiplied out terms of their expansion before simplification. The "P" may remain factorised. Ignore any terms which to powers of x higher than 2.
	A1	For the correct expansion in either form shown. Allow equivalent coefficients. Ignore powers of x higher than 2.
(e)	M1	For an attempt to integrate their expression from part (d), provided it has at least one constant term and at least one algebraic term. See general guidance, no power of x must decrease.
	M1	For substituting in the value of 0.2 into a changed expression and subtracting the correct way. Must see the explicit substitution of 0.2 at least once, if the final answer is not correct. Can be implied if final answer correct. Substitution of 0 does not need to be seen.
	A1	For the value of 0.0093004 [The calculator value is 0.0093316] Accept the value of $\frac{113}{12150}$ if seen.

Type MS · Estimating an integral using a series expansion

Answer 4 of 5 · Question 7 part e · 3 marks

2023 JUN · Paper 02 · extract 3 of 3 · find part e in this mark scheme extract

Question	Scheme	Marks

Type MS · Estimating an integral using a series expansion

Answer 5 of 5 · Question 9 part c · 4 marks

2023 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

9(a)	$\left(1-8x^2\right)^{-\frac{1}{2}}=1+\left(-\frac{1}{2}\right)\left(-8x^2\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-8x^2\right)^2}{2!}$ $+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)\left(-8x^2\right)^3}{3!}$ $\left(1-8x^2\right)^{-\frac{1}{2}}=1+4x^2+24x^4+160x^6+...$	M1 A1A1 [3]
(b)	$\left((a+bx)"\left(1+4x^2+24x^4+160x^6\right)"=\right)\left(a+bx+4ax^2\right)+4bx^3+24ax^4$ $4b=20\Rightarrow b=5$ $24a=48\Rightarrow a=2$	M1A1ft A1 A1 [4]
(c)	$\left(\int_0^{0.2} g(x)\,dx=\int_0^{0.2}\left(\frac{2+5x}{\sqrt{1-8x^2}}\right)dx=\right)$ $\int_0^{0.2}\left("2"+"5"x+"8"x^2+"20"x^3+"48"x^4\right)dx$ $\left(\int_0^{0.2} g(x)\,dx=\right)\left["2x+\frac{5x^2}{2}+\frac{8x^3}{3}+\frac{20x^4}{4}+\frac{48x^5}{5}"\right]_0^{0.2}$ $\left(\int_0^{0.2} g(x)\,dx=\right)"2\times0.2+\frac{5\times0.2^2}{2}+\frac{8\times0.2^3}{3}+\frac{20\times0.2^4}{4}+\frac{48\times0.2^5}{5}"$ $=0.532405333\approx0.5324$ Note: The calculator value is 0.5347698	 M1 M1 M1 A1 [4]
		Total 11 marks

Type MS · Estimating an integral using a series expansion

Answer 5 of 5 · Question 9 part c · 4 marks

2023 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For an attempt at the binomial expansion. The expansion must: • Begin with 1 • The denominators must be correct • The powers of $-8x^2$ must be correct eg $\left(-8x^2\right)^2$ Do not allow missing brackets unless recovered later – this is a general point of marking. Ignore any terms with powers higher than 3. Allow a misread of $\frac{1}{2}$ for $-\frac{1}{2}$ but **not** $8x^2$ for $-8x^2$ - see gen guidance for misreads.
	A1	Following M1 (this is a general point of marking, A marks can only follow M marks). All conditions above met and at least one **algebraic** term correct and simplified.
	A1	Fully correct and simplified
(b)	M1	For attempting to expand $(a+bx)(1+cx^2+dx^4+.....)$ to reach at least the correct terms in x^3 and x^4 for their expansion from part (a) ie $4bx^3$ and $24ax^4$ or cbx^3 and dax^4 Ignore terms with powers higher than 4. It is only necessary to see the correct terms (for their expansion) in x^3 and x^4. They must be using an expansion from (a) of the form $1+cx^2+dx^4+.....$ This mark may be implied from any following working.
	A1ft	For a fully correct expansion in terms of x, a and b,up to the term in x^4 or for $4b = 20$ AND $24a = 48$. **or** if using an incorrect expansion from part (a) of the form $1+cx^2+dx^4+.....$ for a fully correct expansion **or** $cb = 20$ AND $da = 48$. Ignore terms with powers higher than 4.
	A1ft	Either a or b correct ft their expansion from (a) of the form $1+cx^2+dx^4+.....$ **and** $cb = 20$ AND $da = 48$.
	A1ft	Both a and b correct. ft their expansion from (a) of the form $1+cx^2+dx^4+.....$ **and** $cb = 20$ AND $da = 48$.
(c)	M1	For using their expansion from (b) with the correct limits. Although their expression does not need to be fully correct, it must be clear it has arisen from attempting to find a and b in part (b) and using these values. The expression must include the **first** 5 terms in their expansion of the form $p+qx+rx^2+sx^3+tx^4$ from part (b) using their a and b and may include more for this mark. It is not necessary to see the integral sign, if the candidate clearly integrates later. Similarly, if the limits are not present until substitution, this mark can be awarded.
	M1	For an attempt to integrate an expression (see general guidance) – no power of x to decrease. This is not a dependent method mark and the mark is for an attempt to integrate, **but** they must integrate all their terms and there must be a minimum of 2 algebraic terms.
	M1	For substitution of the given limits the correct way around. There must be at least one correct substitution of limits into their **changed** expression with a minimum of 2 algebraic terms (ie this again is not a dependent method mark). 0 need not be substituted if substitution into the changed expression gives an evaluation of 0. This may be implied by a correct answer, but must be shown if the final answer is not correct.
	A1	For awrt 0.5324 ie condone more decimal places.
The question does not demand that students use algebraic integration or show their working clearly and therefore **a correct answer of 0.5324….** may imply the method marks and full marks should be awarded. A value of 0.5347…..will score 0 marks.		

Type MS · Estimating an integral using a series expansion

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Question	Scheme	Marks